

Bubble-Wrap and Box a Fragile Item SOP

This SOP covers packing a single fragile item: wrapping it in two layers of bubble wrap, taping the wrap seam and both open sides, placing it in a box on a foam pad with a second pad on top, applying a fragile sticker, sealing the box and applying the shipping label, using a two-arm robot system. Both arms are used throughout: the grippers cooperate to wrap, tape, box, seal and label, with the left and right grippers taking the specific roles called out in each step. The task runs from an unwrapped item standing at one of three start positions, and an open box at the back-center, to a sealed, stickered and labelled carton standing at the front-center, where it was packed.

The table is set up in one of three ways. The box and the wrap sheet are always in the middle.

- **Config L:** the item is on the left; the pads, tape, sticker, label and waste tray are on the right.
- **Config M:** the item is in the middle, in front of the wrap sheet, which is in front of the box; everything else is on the right.
- **Config R:** the item is on the right; the pads, tape, sticker, label and waste tray are on the left.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the item starts at the back-left (**Config L**), the front-center (**Config M**) or the back-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Supply side:** the pads, tape, sticker, label and waste tray all sit on one side of the table — the right in Config L and M, the left in Config R.
- **Same-side rule:** the gripper on the item's side picks it up and lays it on the sheet. The gripper on the supply side brings every pad, tape strip, sticker and label, and the other gripper takes the other side of it. No arm reaches across the table.
- **Working position:** the item is always moved to the centre of the table, onto the bubble wrap sheet, before any wrapping begins.
- **Wrap target:** every item must be wrapped in two full layers of bubble wrap, bubbles facing the item, with no part of the item showing through the wrap, before it goes into the box.
- **Pack order:** wrap, tape the seam and both sides, box forward, pad base, item, pad top, fragile sticker, seal, label. The order never changes and no step is skipped.
- **Tape rule:** the wrap seam and both open sides of the wrap are taped before the bundle goes in the box; the box seam is taped only after the top pad is in. The longest seam is always taped first.

- **Sticker and label placement:** the fragile sticker goes on the front face of the box and the shipping label goes on the top face, and neither ever overlaps the other or the seal seam.
- **One at a time:** exactly one item, one wrap sheet, one box, two pads, one fragile sticker and one shipping label are handled per episode.

Setup

Go through both checklists before starting the episode.

Hardware checklist

- Cameras are on and recording
- Env camera frame includes the front and back edges of the table and both back corners, so all three item start zones are in frame, and is centered on the table's midpoint
- Both arms are at the home position with grippers open
- Table surface is clear of any objects other than the bubble wrap sheet, the box, the pad stack, the four tape pieces, the fragile sticker, the shipping label, the waste tray and the item

Materials checklist

- One fragile item, the only one handled in the episode, stands in the start zone for this episode's start position, and the other two zones are empty:
 - **Config L:** back-left, clear of the box's folded-out left flap
 - **Config M:** front-center, at the box station, which is empty at the start
 - **Config R:** back-right, where the pad stack normally stands
- One cut bubble wrap sheet is laid flat in the working area at the center, bubble side up, long edge running left to right
- The open box sits at the back-center of the table, at the very back, with all four flaps folded outward and clear of the opening. It is drawn forward in Step 5
- The supplies below are on the right in Config L and M. **IF Config R**, every one of them is in the same spot mirrored to the left: pads at the back-left, tape and waste tray at the left edge, sticker and label at the front-left
- Foam pads are in a single stack at the back-right (at least two pads available)
- Four pieces of tape lie to the right of the working area, at the right edge, each lying free and within reach — three for the wrap and one for the box seam
- One fragile sticker lies face up at the front-right, on its own, ready to be picked
- One shipping label lies face up at the front-right, beside the fragile sticker
- The waste tray is at the right edge and is empty or only part full
- The front-center is clear, so the box has somewhere to come forward to

Workspace layout

- **Item start zone:** the unwrapped fragile item (input) — back-left (Config L), front-center (Config M) or back-right (Config R)
- **Center:** bubble wrap sheet — working area (wrap and tape)
- **Back-center:** the open box at the start of the episode, at the very back
- **Front-center:** box station — where the box is worked once it is brought forward (pad, item, pad, sticker, seal, label), and where the finished carton stays
- **Back-right:** foam pad stack
- **Front-right:** the fragile sticker and the shipping label, lying loose
- **Right edge:** the four tape pieces and the waste tray (wrap offcuts and backing paper only)
- **IF Config R:** the three supply zones above are on the **left** instead — back-left, front-left and left edge

Vocabulary

These are the terms used in this SOP. Operators and annotators must use this language consistently. One term per concept, used throughout.

Item anatomy

- **Item:** the fragile object packed in this task. One item per episode.
- **Rim:** the raised outer edge of the item. Every grasp of the item takes the rim.
- **Face:** the upper surface of the item.
- **Underside:** the lower surface of the item.
- **Contact face:** the underside of the item, the face resting on the wrap sheet at the start.

Packaging anatomy

- **Bubble wrap sheet:** the pre-cut sheet of bubble wrap one item is wrapped in.
- **Bubble side:** the face of the sheet carrying the raised bubbles.
- **Flat side:** the smooth back face of the sheet, opposite the bubble side.
- **Right side of the sheet:** the half of the sheet lying to the right of the item. It goes over the item first.
- **Left side of the sheet:** the half of the sheet lying to the left of the item. It goes over second, on top of the first fold.
- **Layer:** one thickness of the sheet lying over the item. Two folds give two layers.
- **Wrap seam:** the line where the edge of the second fold lands on top of the first fold. It runs the length of the wrapped item and is the longest of the three taped lines.
- **Wrap side:** one of the two open sides of the wrap — the **far side** and the **near side** — left open once both folds are over.

- **Bundle:** the item once it is wrapped in two layers and the seam and both sides are taped.
- **Box:** the open carton the bundle goes into.
- **Box flap:** one of the four top panels of the box that fold over to close it. The two **short flaps** are the left and right ones; the two **long flaps** are the near and far ones. They are told apart by which side of the box they are on, not by their size.
- **Seal seam:** the line where the two long box flaps meet when the box is closed.
- **Foam pad:** one cushioning pad; one goes under the bundle and one on top of it.
- **Pad base:** the foam pad lying flat on the bottom of the box, under the bundle.
- **Pad top:** the foam pad lying flat on top of the bundle, under the flaps.
- **Tape strip:** one of the four pieces of tape lying at the right edge, brought over and pressed down.
- **Fragile sticker:** the adhesive sticker marking the carton as fragile. One per episode, lying loose at the front-right.
- **Shipping label:** the adhesive label carrying the destination details. One per episode, lying loose at the front-right.
- **Carton:** one sealed, stickered and labelled box containing one wrapped item.
- **Box front face:** the vertical face of the box nearest the operator.
- **Box top face:** the upward face of the box once the flaps are closed.

Workspace zones

- **Item start zone:** where the unwrapped item stands at the beginning of the episode — back-left (**Config L**), front-center (**Config M**) or back-right (**Config R**). One per episode.
- **Working area:** the center of the table where the bubble wrap sheet lies and where wrapping and taping happen.
- **Box start position:** the back-center of the table, at the very back, where the open box waits at the beginning of the episode.
- **Box station:** the front-center of the table, where the box is brought forward to and where padding, loading, sealing, stickering and labelling happen. The finished carton stays here.
- **Supply side:** the side of the table holding the pad stack, the tape, the sticker, the label and the waste tray — the right in Config L and M, the left in Config R. The zones below are named for Config L and M; in Config R read left for right.
- **Pad stack:** the foam pad stack at the back of the supply side.
- **Box lane:** the straight path from the box start position at the back-center to the box station at the front-center. The bundle is turned clear of it in Step 4.4 before the box is drawn forward.
- **Label station:** the fragile sticker and the shipping label at the front-right of the table.
- **Tape station:** the four pieces of tape lying at the right edge of the table.

- **Waste tray:** the tray at the right edge that takes wrap offcuts and backing paper and nothing else.
- **Home position:** the default resting pose for each arm: gripper open and clear of the table.

Actions

- **Pick:** lift one item clear of the table and carry it to the working area, with the gripper on the item's side.
- **Turn:** rotate the bundle in place on the sheet, without lifting it off the table.
- **Fold:** bring one side of the wrap sheet over the item and press it down.
- **Bring forward:** draw the box from the box start position at the back-center to the box station at the front-center.
- **Tape:** bring one of the four tape pieces from the right edge and press it down over its full length. Every tape strip in this SOP is applied by both grippers together.
- **Load:** lower the bundle into the box onto the pad base without dropping it.
- **Seal:** press a tape strip along the seal seam so the box stays closed.
- **Apply:** press the fragile sticker or the shipping label flat onto the named box face.

Steps

Step 1 depends on where the item is. Steps 4, 5, 7, 8, 9.2 and 10 depend on the supply side: in Config L and M the **right gripper** brings the pads, tape, sticker and label from the right; in Config R the **left gripper** brings them from the left and the right gripper takes the other side. Steps 2, 3, 6, 9.1 and 11 are the same in all three.

Step 1: Move an item onto the bubble wrap sheet

Goal: one item rests on the bubble wrap sheet in the working area, ready to be wrapped.

Look where the item is before reaching for it.

- **IF the item is at the back-left (Config L):** the **left gripper** grasps it by its **rim**, lifts it clear of the table and carries it forward and to the right, to the center.
- **IF the item is at the front-center (Config M):** the **left gripper** grasps it by its **rim**, lifts it clear of the table and carries it straight back to the center.
- **IF the item is at the back-right (Config R):** the **right gripper** grasps it by its **rim**, lifts it clear of the table and carries it forward and to the left, to the center.

Then, in all three:

- The same gripper lays the item on the bubble wrap sheet, centered, and releases. The other gripper stays clear. The item is never dragged or slid.
- Confirm the bubble side of the sheet is up and there is sheet to the left and to the right of the item.

Step 2: Fold the right side of the sheet over the item

Goal: the right side of the sheet lies over the item as the first layer, bubbles against the item.

- With the **right gripper**, pick the right side of the bubble wrap sheet at its edge and carry it over the item.
- Lay it down over the item and press it against the far side of it, then release.
- Confirm the bubbles face the item and the fold is snug against it with no air gap.

Step 3: Fold the left side over the first fold

Goal: a second layer lies over the item, with its edge pressed down as the wrap seam.

- With the **left gripper**, pinch the left edge of the bubble wrap sheet and carry it over to the right side, over the first wrap.
- Press the edge down onto the wrap to form the **wrap seam**, then release.
- Confirm two full layers cover the item and no part of it shows through the wrap.

Step 4: Tape the wrap seam and both open sides

Goal: the wrap seam and both open sides are taped, so the wrap cannot open on its own, and the bundle lies clear of the box lane.

Start with the longest seam: the wrap seam is taped first, then the far side, then the near side. Every strip is brought by one gripper and applied by both together.

IF Config R: the tape is at the left edge, so in 4.1 to 4.3 the **left gripper** brings each strip and the **right gripper** grasps the other side — the reverse of what is written below.

4.1 Tape the wrap seam

- The **right gripper** grasps one of the tape pieces lying to the right of the workspace and moves it over the wrapped item.
- The **left gripper** then grasps the other side of the tape.
- Both grippers work together to apply it along the longest edge, over its full length, sealing the final fold.

4.2 Tape the far side

- The **right gripper** brings a piece of tape over the bundle and the **left gripper** grasps the other side of it.
- Both grippers work together to press it across the far side, over its full length, so the side is closed.

4.3 Tape the near side

- The **right gripper** brings a piece of tape and the **left gripper** grasps the other side of it, the same way as the far side.

- Both grippers work together to press it across the near side, over its full length, so the side is closed.
- Release both grippers.

4.4 Turn the bundle clear of the box lane

- With the **right gripper** alone, turn the bundle clockwise, in place, until it lies clear of the box lane. The **left gripper** stays clear and does not touch the bundle.
- Release the right gripper.
- Confirm the bundle lies clear of the box lane, so the box can be drawn forward from the back-center to the front-center in Step 5 without touching it.

Warning: Both open sides must be taped before the bundle goes in the box. A taped seam with open sides does not count as wrapped.

Step 5: Bring the box forward and place the pad base

Goal: the box stands at the box station at the front-center, and one foam pad lies flat on the bottom of it, covering it corner to corner.

- Confirm the bundle lies clear of the box lane (Step 4.4) and the box station at the front-center is empty — in Config M it is where the item started.
- **Both grippers work together to bring the box near:** each gripper takes the box at the middle of its own side and they draw it forward together from the back-center to the box station at the front-center, where both arms reach into it. Both grippers release.
- Confirm all four flaps are still folded outward and clear of the opening.
- **IF Config L or M:** with the **right gripper**, grasp the top pad of the **back-right stack**. **IF Config R:** with the **left gripper**, grasp the top pad of the **back-left stack**.
- Lift exactly one pad. If two or more lift together, set them back down and re-grip.
- Lower it into the box and lay it flat on the bottom, then release.
- With either gripper, press the pad down at its center so it sits flat with no lifted corner and no gap at the box walls.

Warning: The box is brought forward before anything goes into it. Nothing is loaded into the box while it is still at the back of the table.

Step 6: Load the bundle into the box

Goal: the bundle sits centered on the pad base, below the top edge of the box.

- With the **right gripper**, grasp the bundle. It is the right gripper alone that lifts the bundle and puts it in the box; the left gripper stays clear.
- Lift the bundle clear of the table, carry it over the box, and lower it onto the pad base in one continuous motion. Do not drop it in.

- Center the bundle in the box.
- Release the right gripper.
- Confirm the top of the bundle sits below the top edge of the box, so the flaps can close over it.

Warning: The bundle must be lowered onto the pad, not dropped. A dropped bundle is a violation even if the item is undamaged.

Step 7: Place the pad top over the bundle

Goal: the second foam pad lies flat on top of the bundle, inside the box and below the flap line.

- **IF Config L or M:** with the **right gripper**, grasp the next pad from the **back-right stack**. **IF Config R:** with the **left gripper**, grasp it from the **back-left stack**. Lift exactly one.
- Lower it into the box onto the bundle and lay it flat, then release.
- Press the pad down at its center so it lies flat and no corner sticks up past the top edge of the box.

Step 8: Apply the fragile sticker

Goal: the fragile sticker is stuck flat on the front face of the box, before the box is sealed.

- With the **right gripper**, take the fragile sticker lying at the front-right and bring it over to the box front face. The **left gripper** takes the other side of it. **IF Config R:** the sticker is at the front-left, so the **left gripper** brings it onto the box front face and the **right gripper** takes the other side.
- Both grippers work together to apply it to the box front face, centered side to side, in the upper third, square to the box, reading the right way up from the operator's side.
- Both grippers press the sticker down over its full area and release.
- If a corner lifts, press it down again. Do not peel the sticker off and re-apply it, and do not add a second sticker.

Step 9: Close the flaps and seal the box

Goal: all four flaps are closed and the seal seam is taped over its full length.

9.1 Close the flaps

- With the left and right grippers, fold the two side flaps in first, one with each gripper: the **right gripper** works on the **right flap** and the **left gripper** works on the **left flap**. Which flap is which is decided by the side of the box it is on, not by its size — some boxes have flaps too alike to tell apart by size.
- Fold the far long flap in with the **left gripper**, then the near long flap in with the **right gripper**, so the two long flaps meet along the seal seam.
- Confirm no pad, wrap or tape is trapped between the flaps.

9.2 Tape the seal seam

- With the **right gripper**, take the last tape piece from the right of the workspace and bring it over the closed flaps.
- The **left gripper** grasps the other side of the strip.
- **IF Config R:** the tape is on the left, so the **left gripper** brings the strip and the **right gripper** grasps the other side.
- Both grippers work together to press it down along the seal seam from one end of the box to the other.
- Release both grippers.

Warning: The box is not sealed until the tape runs the full length of the seal seam. Flaps that are closed but untaped count as unsealed.

Step 10: Apply the shipping label

Goal: the shipping label is stuck flat on the top face of the carton, centered, the right way up, and clear of the seal seam and the fragile sticker.

- With the **right gripper**, take the shipping label lying at the front-right and bring it over to the box top face. The **left gripper** takes the other side of it. **IF Config R:** the label is at the front-left, so the **left gripper** brings it onto the box top face and the **right gripper** takes the other side.
- Both grippers work together to apply it to the box top face, centered on one side of the seal seam, square to the box, reading the right way up from the operator's side.
- Both grippers press the label down over its full area and release.
- Confirm the label does not cross the seal seam and does not overlap the fragile sticker.
- If a corner lifts, press it down again. Do not peel the label off and re-apply it, and do not add a second label.

Step 11: Home the arms and end the episode

Goal: the finished carton stands where it was packed and both arms are back at home.

- Leave the carton at the box station at the front-center, where it was packed. It is not dragged, pushed or lifted to another zone.
- Move both arms back to the home position with grippers open.
- End data collection.

SOP violations

These are the things the operator can do that break (violate) the SOP. This list feeds the violation set and is what reviewers look for using the side-by-side review tool.

How to record a violation in review

For each violation you spot in a recorded episode, record:

- The start timestamp of the violation in the video.
- The violation name from the list below.
- The SOP rule broken (the step number from the list below).

The visible cue is what you actually see in the video. The coaching note is for retraining the operator after the review; it is not what the annotator labels.

Episode handling

Any SOP violation is flagged (or tagged) with the timestamp and violation name. Rather than being discarded, the episode is retained in the training data and tagged with the violation. No flagged episode is deleted. This matches the goal of capturing realistic operator variance in training.

Violations

Note on the start position: the violations below were written for Config L (item starts on the left). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's start position goes under **Config misaligned**.

Violation: Config misaligned.

- **Visible cue:** what the operator does does not match the config on the table — a gripper reaches across the table for the item, a pad, a tape strip, the sticker or the label; the item or the supplies are not where that config puts them; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the same-side rule (the gripper on the item's side picks and lays it; the gripper on the supply side brings every pad, strip, sticker and label; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the item and the supplies are before the first reach, then follow that config's IF lines through Steps 1, 4, 5, 7, 8, 9.2 and 10.

Violation: Wrong pickup.

- **Visible cue:** the item is started from the center or the right instead of the back-left input zone, it is placed outside the bubble wrap sheet, it is grasped anywhere but the rim, or it is dragged or slid across the table instead of being picked up (lifted clear of the surface) and carried.
- **SOP rule broken:** Step 1 (pick from the back-left by the rim and lay it on the wrap sheet). Pick means lift and carry, not drag.
- **Coaching note:** always start each item from the back-left and bring it to the sheet. Take the rim, lift the item clear of the table and carry it. Do not drag or slide it.

Violation: Item laid down off the sheet.

- **Visible cue:** at the end of Step 1 the item sits off the bubble wrap sheet, on its edge, or so far to one side that there is no sheet left to fold over on that side, and the folds start anyway.
- **SOP rule broken:** Step 1 (centered on the sheet, with sheet to the left and to the right of the item).
- **Coaching note:** land the item in the middle of the sheet. A landing off to one side costs a fold.

Violation: Wrapped bubbles-out.

- **Visible cue:** the sheet lies flat side up under the item, so the folds bring the smooth face against the item and the bubbles face outward.
- **SOP rule broken:** the wrap target (bubbles facing the item) and the Setup checklist (sheet laid bubble side up).
- **Coaching note:** check the sheet face before the item goes down. Bubbles always face the item.

Violation: Wrong side folded first.

- **Visible cue:** the left side of the sheet goes over the item before the right side, or one gripper takes both folds.
- **SOP rule broken:** Steps 2 and 3 (right gripper folds the right side first, then the left gripper folds the left side over it).
- **Coaching note:** right side over first with the right gripper, left side over second with the left gripper. The seam only lands right if the folds go in that order.

Violation: Fewer than two layers of wrap.

- **Visible cue:** the item is covered by only one layer, or the second fold stops short and part of the first fold is still exposed at the end of Step 3.
- **SOP rule broken:** Step 3 (two full layers over the item).
- **Coaching note:** two full folds, every time. If the second side comes up short, open the wrap out and lay the item down again with more sheet on the left.

Violation: Item visible through the wrap.

- **Visible cue:** at the end of Step 3 part of the item is uncovered — the rim or part of the face showing at an edge of the wrap.
- **SOP rule broken:** Step 3 (no part of the item shows through the wrap).
- **Coaching note:** walk the wrap around the whole item and check the rim all the way round before taping.

Violation: Wrap seam not taped.

- **Visible cue:** the edge of the second fold is pressed down but no tape strip is run along the wrap seam, or the tape is pressed at one point only.
- **SOP rule broken:** Step 4.1 (both grippers press one tape strip along the wrap seam over its full length).
- **Coaching note:** run the strip end to end along the seam. A spot press does not hold the wrap.

Violation: Tape applied with one gripper.

- **Visible cue:** one gripper brings a strip and lays it down on its own, without the other gripper taking the far side of the strip, on the wrap seam, either wrap side, or the seal seam.
- **SOP rule broken:** Steps 4.1, 4.2, 4.3 and 9.2 (one gripper brings the tape, the other takes the other side, both apply it together).
- **Coaching note:** tape is a two-gripper job here. Hand the far side over before pressing.

Violation: Seams taped out of order.

- **Visible cue:** a wrap side is taped before the wrap seam, or the near side is taped before the far side.
- **SOP rule broken:** Step 4 (longest seam first, then the far side, then the near side).
- **Coaching note:** longest seam first, then far, then near. The long seam is what holds the wrap shut.

Violation: Wrap side not taped.

- **Visible cue:** one or both wrap sides are left untaped, or the strip is pressed at one point only instead of across the side.
- **SOP rule broken:** Steps 4.2 and 4.3 (tape each wrap side closed over the full length of the strip).
- **Coaching note:** do both sides, end to end. Open sides let the item slide out of the wrap.

Violation: Bundle boxed before the seam and both sides are taped.

- **Visible cue:** the bundle goes into the box while the wrap seam or either wrap side is still untaped.
- **SOP rule broken:** Step 4 (wrap seam and both wrap sides taped before the bundle goes in the box).
- **Coaching note:** finish all three tape strips at the working area. Nothing enters the box half taped.

Violation: Box loaded before it is brought forward.

- **Visible cue:** the pad base, the bundle or the pad top goes into the box while the box is still at the back-center, or the box is never drawn forward to the box station at all.

- **SOP rule broken:** Step 5 (draw the box forward to the front-center before anything goes into it).
- **Coaching note:** bring the box to you first. Reaching to the back of the table to load it is what tips the bundle.

Violation: Pad base missing or not flat.

- **Visible cue:** the bundle is loaded onto the bare bottom of the box, or the pad base is left folded, crooked, or with a lifted corner or a gap at the box walls.
- **SOP rule broken:** Step 5 (one pad flat on the bottom of the box, corner to corner, pressed down).
- **Coaching note:** the pad goes in first and gets pressed flat. Check for lifted corners before the bundle goes in.

Violation: More than one pad lifted.

- **Visible cue:** two or more pads come off the back-right stack together and the operator continues without setting them back down and re-gripping.
- **SOP rule broken:** Steps 5 and 7 (lift exactly one pad; if two or more lift together, set them back down and re-grip).
- **Coaching note:** separate the top pad from the stack before lifting, and put a double pick straight back down.

Violation: Bundle dropped into the box.

- **Visible cue:** the bundle is released above the box and falls onto the pad base rather than being lowered onto it in a continuous motion.
- **SOP rule broken:** Step 6 (lower the bundle onto the pad base; do not drop it in).
- **Coaching note:** carry the bundle all the way down to the pad before opening the gripper. This is a fragile item.

Violation: Bundle off-center or proud of the box.

- **Visible cue:** at the end of Step 6 the bundle sits on top of the pad edge, or stands above the top edge of the box so the flaps cannot close.
- **SOP rule broken:** Step 6 (centered on the pad base, below the top edge of the box).
- **Coaching note:** center the bundle and check the height before the top pad goes in.

Violation: Pad top missing or not flat.

- **Visible cue:** the box is closed with no pad on top of the bundle, or the top pad is folded, crooked, or sticking up past the top edge of the box.
- **SOP rule broken:** Step 7 (one pad flat on top of the bundle, below the flap line, pressed down).

- **Coaching note:** the second pad is not optional. Lay it flat and press it down before closing the flaps.

Violation: Fragile sticker missing, crooked, or on the wrong face.

- **Visible cue:** the carton leaves Step 8 with no fragile sticker, with the sticker on a face other than the front face, upside down, or visibly off center or off square on that face.
- **SOP rule broken:** Step 8 (one fragile sticker, box front face, centered in the upper third, square to the box, reading the right way up).
- **Coaching note:** aim the sticker before releasing it: front face, upper third, centered, right way up.

Violation: Sticker or label peeled off and re-applied, or doubled.

- **Visible cue:** a placed sticker or label is pulled back off and stuck down again, or a second one is added over or beside the first.
- **SOP rule broken:** Steps 8 and 10 (press a lifted corner back down; do not peel and re-apply, and do not add a second sticker or label).
- **Coaching note:** a lifted corner gets pressed, not removed. One sticker and one label per carton.

Violation: Flaps closed in the wrong order or with something trapped.

- **Visible cue:** the long flaps are folded in before the side flaps, the near long flap goes in before the far one, a gripper works the flap on the other side of the box (the left gripper on the right side flap or the near long flap, the right gripper on the left side flap or the far long flap), or a pad, the wrap, or a tape end is trapped between the flaps at the end of Step 9.1.
- **SOP rule broken:** Step 9.1 (side flaps first, one with each gripper, then the far long flap with the left gripper, then the near long flap with the right gripper, with nothing trapped between them).
- **Coaching note:** side flaps first, far long flap next, near long flap last, and clear the flap line before the long flaps come over. The flap is picked by which side of the box it is on, not by how big it looks.

Violation: Box closed but not sealed.

- **Visible cue:** the flaps are folded over and released without a tape strip run along the seal seam, or the tape covers only part of the seam.
- **SOP rule broken:** Step 9.2 (both grippers press one tape strip along the seal seam from one end of the box to the other).
- **Coaching note:** run the tape the full length of the seam. Closed flaps are not a seal.

Violation: Shipping label missing, crooked, or on the wrong face.

- **Visible cue:** the carton leaves Step 10 with no shipping label, with the label on a face other than the top face, upside down, or visibly off center or off square on that face.
- **SOP rule broken:** Step 10 (one label, box top face, centered on one side of the seal seam, square to the box, reading the right way up).
- **Coaching note:** aim the label before releasing it: top face, one side of the seam, centered, right way up.

Violation: Label crossing the seal seam or overlapping the sticker.

- **Visible cue:** the shipping label is placed across the taped seal seam, or it overlaps the fragile sticker.
- **SOP rule broken:** Step 10 (the label does not cross the seal seam and does not overlap the fragile sticker).
- **Coaching note:** place the label on one side of the seam, clear of the sticker. A label across the seam tears when the box is opened.

Violation: Finished carton moved out of the box station.

- **Visible cue:** after the label is applied, the carton is dragged, pushed or lifted away from the front-center where it was packed, to the back of the table or off to one side.
- **SOP rule broken:** Step 11 (the carton stays at the box station where it was packed).
- **Coaching note:** the carton is finished where it stands. Nothing moves it after the label goes on.

Violation: Sequence out of order.

- **Visible cue:** the operator seals the box before the top pad is in, applies the fragile sticker after sealing, tapes the box seam before the wrap sides, or otherwise departs from the pack order.
- **SOP rule broken:** the pack order (wrap, tape the seam and both sides, box forward, pad base, item, pad top, fragile sticker, seal, label).
- **Coaching note:** follow the pack order every episode. Each step depends on the one before it.

Violation: Waste not placed in the waste tray.

- **Visible cue:** a wrap offcut or a piece of backing paper is dropped on the table, left in the working area, or another object is put into the waste tray.
- **SOP rule broken:** the waste tray rule (offcuts and backing paper go to the tray at the right edge; nothing else goes in it).
- **Coaching note:** carry the offcut all the way to the tray and release it there. The tray takes offcuts and backing paper only.

Violation: Wrong arm used for an action.

- **Visible cue:** any step that specifies the left gripper or the right gripper is performed with the opposite gripper, a step that calls for both grippers together is done with one, or the left gripper lifts the bundle into the box in Step 6.
- **SOP rule broken:** any step that specifies a gripper, including Steps 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10.
- **Coaching note:** operator confusion about left vs right roles. Walk through the SOP step by step with the operator.

Violation: Re-grip on a pick or grasp.

- **Visible cue:** the operator closes on the item, the sheet, the box, a pad, the tape, the sticker or the label, finds the grip off, opens, and re-grips before lifting more than two times.
- **SOP rule broken:** Steps 1/2/3/4/5/6/7/8/10 (grasp securely, e.g., the rim, the sheet edge, the box edge, the pad, a tape piece or the sticker/label).
- **Coaching note:** approach angle off. Practice the from-above approach so the first grip catches the right point.

Violation: Repeated fiddling with placement, wrap or tape.

- **Visible cue:** the operator makes many small adjustments (more than about two) to settle the item, the wrap, the box, a pad, a tape strip, the sticker or the label, rather than settling it in one or two corrections.
- **SOP rule broken:** the fold/tape/load/apply sub-steps (settle in one or two small adjustments).
- **Coaching note:** grip or approach is off, forcing repeated correction. Tighten the approach so each fold, each strip and each placement lands close to target.

Violation: Arms not fully at home position at episode end.

- **Visible cue:** in the final frame, both arms are close to the home position but not exactly at it (gripper not fully open, or arm position visibly off home).
- **SOP rule broken:** Step 11 (return to home position with grippers open).
- **Coaching note:** complete the home motion explicitly before ending recording.

Non-violation failures

These are episode failures that are not the operator's fault and do not go in the violation set. They are recorded as system issues, the episode is discarded/deleted, and the episode does not become a coaching point for the operator.

- **Recording stopped or paused mid-episode:** Cause: software or hardware issue with the recording system.
- **Camera dropped frames or lost feed during the episode:** Cause: camera or capture system issue.

- **Hardware fault on the robot arm:** Cause: gripper malfunction, arm position drift, or motor error during the episode.
- **Defective item:** Cause: an item that arrives already chipped, cracked or broken, through no fault of the operator. Replace before the next episode.
- **Defective bubble wrap sheet:** Cause: a sheet cut too short for two layers, or one with the bubbles already crushed flat. Replace before the next episode.
- **Defective box or pads:** Cause: a crushed or split box, flaps that will not hold a fold, or pads torn or too small for the box footprint. Replace before the next episode.
- **Defective tape:** Cause: a piece of tape that is torn, stuck to itself, or has lost its adhesive. Replace the piece before the next episode.
- **Defective sticker or label:** Cause: a sticker or label that has lost its adhesive or is torn. Replace before the next episode.

After each episode: repeat or reset

- **Batch sessions (e.g., 5x):** if fewer than the target number are packed, reset the workspace and return to Step 1 with the same item, once the reset has stood it back in its start zone. Once the target number is packed, reset the workspace before the next session.
- Clear the working area and re-run the Setup checklist before the next session.

After the episode: reset the workspace

This part is not recorded. It is just how you reset the table for the next episode.

- With recording off, take the finished carton from the box station at the front-center, where it was packed.
- Cut the carton open, take the bundle out, unwrap the item, and stand it in the start zone for the next episode's start position, leaving the other two zones empty. If the config changes to or from Config R, move the pads, tape, sticker, label and waste tray to the other side.
- Discard the used wrap, tape, stickers and labels, and set the used pads aside if they are still flat and undamaged.
- Restock the pad stack at the back-right, and lay one fresh fragile sticker and one fresh shipping label face up at the front-right.
- Lay out four fresh pieces of tape to the right of the working area.
- Lay a fresh cut bubble wrap sheet flat in the center, bubble side up, long edge running left to right.
- Set a fresh open box at the back-center, at the very back of the table, with all four flaps folded outward and clear of the opening, and leave the front-center clear for it to come forward to.
- Empty the waste tray at the right edge.

- Confirm the table surface is clear of any objects other than the wrap sheet, the box, the pad stack, the four tape pieces, the fragile sticker, the shipping label, the waste tray and the item in their designated zones.
- Go through the Setup checklist again before starting the next episode.

Warning: The table must be clear except for the bubble wrap sheet in the center, the open box at the back-center, the pad stack at the back-right, the fragile sticker and the shipping label at the front-right, the four tape pieces and the waste tray at the right edge, and the item in its start zone.

Annotation subtasks

1. Move one item from its start zone onto the bubble wrap sheet
2. Fold the right side of the sheet over the item
3. Fold the left side of the sheet over the first fold
4. Tape the wrap seam
5. Tape the far side of the wrap
6. Tape the near side of the wrap
7. Bring the box forward and place the pad base flat in it
8. Load the bundle onto the pad base
9. Place the pad top flat over the bundle
10. Apply the fragile sticker to the box front face
11. Close the flaps and tape the seal seam
12. Apply the shipping label to the box top face
13. Home the arms and end the episode